

Module 07: Communication — Neural Resonance, Mirror Systems, and Social Cognition

Learning Objectives

1. Describe how the **mirror neuron system** supports action understanding and social inference through motor simulation.
2. Explain neural resonance and interpersonal synchrony as the neural basis of coupled Active Inference between agents.
3. Analyze the neural underpinnings of Theory of Mind, including the temporo-parietal junction and medial prefrontal cortex.
4. Connect communicative inference to **pragmatics** and Bayesian models of language understanding.

Introduction

Communication requires that isolated brains — each behind its own skull — achieve sufficient shared understanding to coordinate behavior. How does this happen neurally? This module examines three interacting systems: the **mirror neuron system** (which maps observed actions onto the observer's own motor repertoire), the **mentalizing network** (which represents other agents' beliefs and intentions), and the phenomenon of **interpersonal neural synchrony** (where interacting brains entrain to each other).

Key Concepts

1. Mirror Neurons and Motor Simulation

Mirror neurons, discovered by Rizzolatti et al. (1996) in premotor cortex of macaques, fire both when an animal performs an action and when it observes the same action performed by another. In Active Inference terms, mirror neurons implement **motor simulation** — the observer generates the same proprioceptive predictions as the actor, allowing the observer's generative model to "predict" the actor's intentions by simulating them internally.

In humans, the mirror neuron system (inferior frontal gyrus, inferior parietal lobule) provides the sensorimotor basis for understanding others' actions without explicit propositional reasoning.

2. Interpersonal Neural Synchrony

Hyperscanning studies (simultaneous recording of brain activity from multiple interacting individuals) reveal that interacting brains show **neural synchrony** — correlated oscillatory activity, particularly in the alpha and gamma bands. Hasson et al. (2012) demonstrated speaker-listener neural coupling: the listener's brain activity patterns converge with the speaker's, with a temporal lag corresponding to comprehension. Remarkably, successful communication correlated with tighter coupling, while miscomprehension correlated with decoupling.

In Active Inference terms, interpersonal synchrony is the neural signature of **generalized synchrony** — the two agents' generative models aligning through reciprocal prediction error exchange.

3. The Mentalizing Network

Theory of Mind — the capacity to attribute beliefs and intentions to others — depends on a distinct neural network: the **temporo-parietal junction (TPJ)**, **medial prefrontal cortex (mPFC)**, **posterior superior temporal sulcus (pSTS)**, and **temporal poles**. Saxe and Kanwisher (2003) showed that the TPJ is selectively activated when participants reason about others' beliefs, not simply about physical events.

In Active Inference terms, the mentalizing network implements the hierarchical component of the generative model that represents *other agents as inference systems* — modeling not just what another person does but what they believe, desire, and intend.

4. Pragmatics and Communicative Inference

Language understanding goes far beyond decoding words. **Pragmatics** — the study of how context shapes meaning — reveals that comprehension requires inference about the speaker's

communicative intention. Grice's (1975) cooperative principle and its maxims (quantity, quality, relevance, manner) describe the shared expectations that make inference tractable.

Goodman and Frank (2016) formalized this as the **Rational Speech Act** (RSA) model: the listener infers what the speaker meant by reasoning about what a rational speaker would say given their communicative goals. This is Bayesian inference over communicative intentions — a direct application of Active Inference to language. The listener's generative model includes a model of the speaker's generative model — a hierarchy of inference over inference.

5. Language as Predictive Processing

Language comprehension engages the same predictive machinery as perception. Kutas and Hillyard (1980) discovered the **N400** event-related potential — a neural signature of semantic prediction error. Unexpected words ("He spread the warm bread with *socks*") produce a large N400; expected words ("...with *butter*") produce a small N400. This demonstrates that the brain continuously generates predictions about upcoming words during sentence processing.

Kuperberg and Jaeger (2016) extended this to a full Active Inference account of language: syntactic and semantic predictions are generated hierarchically and updated through prediction error minimization, with precision weighting determining whether bottom-up input or top-down expectations dominate comprehension.

Clinical Connections

- **Autism and social communication:** Some accounts propose reduced spontaneous mirror activity or reduced mentalizing precision, leading to difficulties in automatic social inference (though compensatory explicit reasoning may develop).
- **Schizophrenia and paranoia:** Overactive mentalizing (higher-order ToM) may lead to aberrant attribution of intentions to others — e.g., perceiving neutral actions as threatening or conspiratorial.
- **Social anxiety:** Excessive precision on social prediction errors — the brain over-attends to signs of social rejection or judgment, leading to avoidance behavior.

Conclusion

Social communication is built on a neural infrastructure that implements coupled inference between brains — from the low-level motor resonance of the mirror system to the high-level belief modeling of the mentalizing network, and extending into the pragmatic inference that makes language meaningful. Module 08 examines the temporal dimension of social and individual cognition — planning, decision-making, and the prefrontal cortex.

Further Reading

- Rizzolatti, G. et al. (1996). Premotor cortex and the recognition of motor actions. *Cognitive Brain Research*, 3(2), 131-141.
- Hasson, U. et al. (2012). Brain-to-brain coupling: A mechanism for creating and sharing a social world. *Trends in Cognitive Sciences*, 16(2), 114-121.
- Goodman, N. D. & Frank, M. C. (2016). Pragmatic language interpretation as probabilistic inference. *Trends in Cognitive Sciences*, 20(11), 818-829.